When an application cannot be supported by the 8 bit ANSEL character set, one can choose character set(s) from those registered by the Data Administration Section of the Family History Department. Those character sets which are currently registered are found in Appendix A. Those who have need to use character sets other than those registered, please consult the Family History Department Data Administration Section.

Changing Character Sets

The default and initial character set for a GEDCOM transmission is the 8 bit ANSEL character set.

If the data to be interchanged is in a character set other than the ANSEL, or when the data changes from one character set to another, the change is to be indicated by a character set change escape sequence.

The character set change escape sequence is to be represented by the following conventions:

(space)@#C…:LEN@

The (space)@#C signals a change from the present character set. The ... represents a code designating what new character set (or non-character representation) will be in effect as registered in Appendix A. The :LEN is an optional attribute (variable length) that states the length, in bits for non-character data [such as binary data] or number of characters in the new character set, of the following changed data. At the end of that length, representation returns to the current character set. The trailing @ signals that the next byte or bit begins the new representation.

The escape sequence is context insensitive and may appear any place and as many times as needed in the body of the data, including within a given field. This sequence is ignored in identifying the parts of a GEDCOM line, such as level, tag value, and terminator.

The occurrence of the character set escape sequence signals that the data following the escape sequence is represented in a different character set, as shown here:

XXXXXXXXXXX @#C002@YYYYYYYYYYY @#C001@XXXXXXXX

where

- X indicates a character in the 8 bit ANSEL character set,
- Y indicates a character in 8 bit US ASCII, and

Note that the at sign, pound sign, the letter C, and the three digits of the character set code will be represented by different character codes in each case, depending on the codes assigned for those symbols in each character set. The escape sequence is given in character codes from the most recently-specified character set.

Note also that all three changes shown above could have occurred on the same line.

APPENDIX A. GEDCOM Character Sets

The following codes are placed in a GEDCOM line to indicate a change to the associated character set:

@#C001 (Default)	8 bit ANSEL character set
@#C002	US ASCII (ANSI 8-bit)

APPENDIX B. 8 bit ANSEL Character Set

ANSEL character set pages 6 – 10

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		0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111	8765 BITS
		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
0000	0	NUL	DLE	SP	0	@	P	` *	P				ʿ	°		˺	˻	
0001	1	SOH	DC1	!	1	A	Q	a	q			Ł	ł	ℓ		ˋ	ˏ	
0010	2	STX	DC2	"	2	B	R	b	r			Ø	ø	℗		´	ˎ	
0011	3	ETX	DC3	#	3	C	S	c	s			Đ	đ	©		ˆ	ˍ	
0100	4	EOT	DC4	$	4	D	T	d	t			Þ	þ	♯		˜	˚	
0101	5	ENQ	NAK	%	5	E	U	e	u			Æ	æ	¿		¯	˝	
0110	6	ACK	SYN	&	6	F	V	f	v			Œ	œ	¡		˘	—	
0111	7	BEL	ETB	´	7	G	W	g	w			'	"			˙	˛	
1000	8	BS	CAN	(	8	H	X	h	x			·	ı			¨	˹	
1001	9	HT	EM	)	9	I	Y	i	y			þ	£			˚	˼	
1010	10	LF	SUB	*	:	J	Z	j	z			®	ð			°	˽	
1011	11	VT	ESC	+	;	K	[	k	{			±				˾	˿	
1100	12	FF	FS	,	<	L	\	l	|			Ơ	ơ			˼		
1101	13	CR	GS	-	=	M	]	m	}			Ư	ư			˛		
1110	14	SO	RS	.	>	N	^ *	n	~ *			'				˝	˛	
1111	15	SI	US	/	?	O	_ *	o	DEL							˚		
4321 BITS																		

*Redefined in the extended Latin alphabet coded character set.

Reserved for control characters.

Reserved for future standardization.

Corners (Reserved).

6. Legend

NOTE: The legend for ASCII character in 6.3 and 6.4 is adapted from ANSI X3.4-1977. For complete information of the use of ASCII refer to the latest edition of that standard.

6.1 ANSEL Spacing Graphic Characters

7-Bit Col/Row	8-Bit Col/Row	Graphic	Name	Example of Use
2/1	10/1	Ł	slash L - uppercase	Łódź
2/2	10/2	Ø	slash O - uppercase	Øst
2/3	10/3	Đ	slash D - uppercase	Đuro
2/4	10/4	Þ	thorn - uppercase	Þann
2/5	10/5	Æ	ligature AE - uppercase	Ægir
2/6	10/6	Œ	ligature OE - uppercase	Œuvre
2/7	10/7	′	mi͡agkiĭ znak	Fakul′tet
2/8	10/8	·	middle dot	novel·la
2/9	10/9	♭	musical flat	B♭
2/10	10/10	®	patent mark	ABC®
2/11	10/11	±	plus or minus	A±B
2/12	10/12	Ȯ	hook O - uppercase	BȮ
2/13	10/13	Ư	hook U - uppercase	XƯA
2/14	10/14	ʼ	alif	Unʼyusho
3/0	11/0	ʻ	ʻayn	faʻil
3/1	11/1	ł	slash l - lowercase	rozbił
3/2	11/2	ø	slash o - lowercase	Høj
3/3	11/3	đ	slash d - lowercase	đavola
3/4	11/4	þ	thorn - lowercase	Þann
3/5	11/5	æ	ligature ae - lowercase	skæg
3/6	11/6	œ	ligature oe - lowercase	œuvre
3/7	11/7	″	tvërdyĭ znak	ob″i͡avlenie
3/8	11/8	ı	dotless i - lowercase	masalı
3/9	11/9	£	British pound	£5.00
3/10	11/10	ð	eth	verður
3/12	11/12	ơ	hook o - lowercase	Sở
3/13	11/13	ư	hook u - lowercase	Tư Dưc
4/0	12/0	°	degree sign	10°C
4/1	12/1	ℓ	script l[2]	25 ℓ
4/2	12/2	℗	phonograph record copyright mark	Decca ℗
4/3	12/3	©	copyright symbol	© 1974
4/4	12/4	♯	musical sharp	D♯
4/5	12/5	¿	inverted question mark	¿Qué?
4/6	12/6	¡	inverted exclamation mark	¡Esta!

[2] In bibliographic work, the script l, ℓ, is commonly used as an abbreviation for the term “leaves.” It shall not be used as a symbol for the unit of measure “liter.”

6.2 ANSEL Nonspacing Graphic Characters

7-Bit Col/Row	8-Bit Col/Row	Graphic	Name	Example of Use
6/0	14/0	̏	low rising tone mark	củi
6/1	14/1	̀	grave accent	règle
6/2	14/2	́	acute accent	está
6/3	14/3	̂	circumflex accent	même
6/4	14/4	̃	tilde	niño
6/5	14/5	̄	macron	gājējs
6/6	14/6	̆	breve	altă
6/7	14/7	̇	dot above	żaba
6/8	14/8	̈	umlaut (diaeresis)	öppna
6/9	14/9	̌	háček	vždy
6/10	14/10	̊	circle above (angstrom)	hår
6/11	14/11	͡	ligature, left half	akademii͡a
6/12	14/12	͜	ligature, right half	akademii͡a
6/13	14/13	̓	high comma, off center	rozdel̓ovac
6/14	14/14	̋	double acute accent	időszaki
6/15	14/15	̐	candrabindu	Aliĭev
7/0	15/0	̧	cedilla	ça
7/1	15/1	̨	right hook	vietą
7/2	15/2	̣	dot below	teḍa
7/3	15/3	̤	double dot below	khuṭbah
7/4	15/4	̥	circle below	Maharṣicāritamṛtam
7/5	15/5	̳	double underscore	G̳hulam
7/6	15/6	̲	underscore	samar
7/7	15/7	̦	left hook	darziņa
7/8	15/8	̩	right cedilla	khọng
7/9	15/9	̪	half circle below (upadhmaniya)	ḫumantuš
7/10	15/10	͠	double tilde, left half	͠ngalan
7/11	15/11	͡	double tilde, right half	͡ngalan
7/14	15/14	̕	high comma, centered	ġeotermika

6.3 ASCII Control Characters

Col/Row	Mnemonic and Meaning		Col/Row	Mnemonic and Meaning	
0/0	NUL	Null	1/0	DLE	Data Link Escape
0/1	SOH	Start of Heading	1/1	DC1	Device Control 1
0/2	STX	Start of Text	1/2	DC2	Device Control 2
0/3	ETX	End of Text	1/3	DC3	Device Control 3
0/4	EOT	End of Transmission	1/4	DC4	Device Control 4
0/5	ENQ	Enquiry	1/5	NAK	Negative Acknowledge
0/6	ACK	Acknowledge	1/6	SYN	Synchronous Idle
0/7	BEL	Bell	1/7	ETB	End of Transmission Block
0/8	BS	Backspace	1/8	CAN	Cancel
0/9	HT	Horizontal Tabulation	1/9	EM	End of Medium
0/10	LF	Line Feed	1/10	SUB	Substitute
0/11	VT	Vertical Tabulation	1/11	ESC	Escape
0/12	FF	Form Feed	1/12	FS	File Separator
0/13	CR	Carriage Return	1/13	GS	Group Separator
0/14	SO	Shift Out	1/14	RS	Record Separator
0/15	SI	Shift In	1/15	US	Unit Separator
			7/15	DEL	Delete

6.4 ASCII Graphic Characters

Col/Row	Graphic	Name
2/0	SP	Space (Normally Nonprinting)
2/1	!	Exclamation Point
2/2	"	Quotation Marks (Diaeresis)
2/3	#	Number Sign
2/4	$	Dollar Sign
2/5	%	Percent Sign
2/6	&	Ampersand
2/7	´	Apostrophe (Closing Single Quotation Mark; Acute Accent)
2/8	(	Opening Parenthesis
2/9	)	Closing Parenthesis
2/10	*	Asterisk
2/11	+	Plus
2/12	,	Comma (Cedilla)
2/13	-	Hyphen (Minus)
2/14	.	Period (Decimal Point)
2/15	/	Slant
3/0 to 3/9	0 … 9	Digits 0 through 9
3/10	:	Colon
3/11	;	Semicolon
3/12	<	Less Than
3/13	=	Equals
3/14	>	Greater Than
3/15	?	Question Mark
4/0	@	Commercial At
4/1 to 5/10	A … Z	Uppercase Latin Letters A through Z
5/11	[	Opening Bracket
5/12	\	Reverse Slant
5/13	]	Closing Bracket
5/14	^	Circumflex
5/15	_	Underline
6/0	`	Opening Single Quotation Mark (Grave Accent)
6/1 to 7/10	a … z	Lowercase Latin Letters a through z
7/11	{	Opening Brace
7/12	|	Vertical Line
7/13	}	Closing Brace
7/14	~	Tilde

Approved: Version 1.1, 9 October 1987 Number: 0046e

INTRODUCTION

The GEDCOM format is used to transfer a wide variety of data to a wide variety of computer systems. It is necessary to provide identifying and routing information in header and trailer records on each GEDCOM transmission to make sure the data gets to the right place. This section describes the tags and values required in GEDCOM header and trailer records.

SCOPE

This standard specifies GEDCOM transmission header and trailer formats. The header and trailer are used by the Family History Departments Receiving and Routing System to insure that are correctly routed. These formats may also include information to help an application software system process the data in the transmission. The header and trailer are separate from any other protocols attached to the transmission by an electronic communications network.

This standard does not address the content of application messages, character sets used to represent data, or transmission error correction.

APPLICABILITY

This standard applies to GEDCOM transmissions among all Family History Department applications and to transmissions among other applications that interface with Family History Department applications.

STANDARD

Transmission Header

A transmission header has both required and optional GEDCOM lines. Each Line begins with a level number and a tag (capital letters), and may be followed by a value. The optional lines identified in this standard do not constitute an exhaustive list of optional lines. They simply represent some possibilities.

Required Lines

The following lines are required in all GEDCOM transmission headers.

- HEAD. The first line of a GEDCOM transmission is always a HEAD line occurring at level 0. No value is specified. All other GEDCOM lines in the header are at levels subordinate to the header.

- 1 SOUR. The content of the value field for this SOURce line is the resource identifier (the name) of the system or file where the transmission originates and the version number of the system. This appears as the system identifier, a space, then the version number.

- 1 DEST. The value for this DESTination line is the resource identifier of the system or file to which the transmission is being sent. If the transmission is to be sent to multiple destinations, include a separate DEST line with resource identifier for each destination. Do not include a version number, it is not required or needed.

A resource identifier is a single string of alphanumeric characters that identify any system, file, etc. that may participate in a GEDCOM transmission as either a sender or a receiver. The receiving and routing function of the Family History Department associates this identifier with the actual locations of the sender and receiver. This permits relocation of systems and files without modification of the systems that initiate transmissions or of data containing pointers to records in other systems. The resource identifier is assigned by the Data Administration section of the Family History Department.

Optional Lines

- 1 DATE. This line has the date the GEDCOM transmission was generated by the sender. The content of the value field for this line is discussed in the STANDARD FOR GEDCOM VALUE FORMATS.

- 1 TIME. This line has the time the GEDCOM transmission was generated by the sender. content of the value field for this line is discussed in the STANDARD FOR GEDCOM VALUE FORMATS.

- 1 CHAR. This line is used to change from the default character set of the header to some other character set. If used, it must follow the required lines of the header.

Character Set

The required lines of a GEDCOM transmission are always coded in 8-bit ANSEL characters restricted to decimal codes 10 (line feed), 13 (carriage return), and 32 thru 126 (printable characters). The characters specified are identical in the 7-bit US ASCII character set.

If the computer generating the transmission cannot accommodate this default character set, at least the required lines of the header must be converted to it before sending the transmission to any other computer which does support it.

A change of character sets, whether in the header or in the body of the data, has effect only for that transmission. Any subsequent transmissions must begin in the default character set.

Transmission Trailer

A GEDCOM transmission is terminated by a single TRLR (trailer) line occurring at level 0. No value field is specified for this line.

Approved: Version 1.1, 9 October 1987 Number: 0047e

INTRODUCTION

This standard is needed in order to establish the physical environment in which GEDCOM is used.

SCOPE

This standard specifies what physical media may be used to transport a GEDCOM transmission from one computer to another computer and specifies how the media itself is physically organized to carry the data. This standard does not address the content of application messages or character sets used to represent data.

APPLICABILITY

This standard applies to interfaces between ail Family History Department data processing applications using the GEDCOM format and to other applications which interface with Family History Department applications using the GEDCOM format.

It also applies to computer media which are used to archive data from. Family History Department applications in GEDCOM format.

It does not specifically constrain the transfer of data between tasks within an application, but should be applied when such transfers are done using GEDCOM.

REFERENCES

Information Systems Practice 202. Data Communications Standards
Information Systems Practice 203. Information Systems Hardware and System Software Product Shopping List Standard

DEFINITIONS

Electronic Communications. The means.to transfer information between two separate computer systems via a direct, electrical connection between them.

STANDARD

3 1/2" Floppy Diskette

720K formatted capacity diskette constructed and formatted for use in the IBM PC Convertible lap top computer running MS DOS 3.2 and -the IBM PS2 computer. The 1.44M formatted diskette for the IBM PS2 is also an acceptable diskette for use in this standard.

400K and 800K formatted Apple Macintosh diskette.

5 1/4" Floppy Diskette

- 360K formatted capacity diskette constructed and formatted for use in the IBM PC standard desk top computer running MS DOS 2.1.

- 1.2M formatted capacity diskette constructed and formatted for use in the IBM PC AT desk top computer running MS DOS 3.1.

- 124K (16 sector) formatted capacity diskette constructed and formatted for use in the Apple II family of computers running Apple DOS 3.3.

- 191K single-sided, double density capacity diskette constructed and formatted for use in the Kaypro family of computers running CP/M.

5 1/4" Iomega 20 Megabyte Cartridge

1/2" Magnetic Tape

9-track, 1600 BPI, unlabeled, phase encoded magnetic tape or 9-track, 6250 BPI, labeled, Group Coded Recording (GCR) magnetic tape.

Electronic Communications

Standards for electronic communications as specified in the ISD's Information Systems Practices 202 and 203 will be followed. For PC to PC type asynchronous communications over dial-up phone lines, these standards call for the use of Bell 212A type models at 300 or 1200 bps with either TTY, Hayes Verification, XMODEM, or X.PC protocols.

Internal Disk Files

The normal internal text file format of the computer in which the GEDCOM data is placed.

Approved: Version 0.3, 9 October 1987 Number: 0048e

INTRODUCTION

A common understanding of what tags mean is required in order to retain uniformity in the identification of information that is being transmitted. This document has been prepared to give a basis for that common definition of GEDCOM tags.

SCOPE

All GEDCOM tags that are being widely used and require a common understanding and definition are to be included as a part of this document.

Tags that do not conform to the current requirements for data tags as outlined in the document which contains the standards for GEDCOM data tags are not included in this document.

APPLICATION

The GEDCOM data tags that have been approved and defined are in this document and should be used in accordance with the definition that accompanies the tag. If you require that a new data tag be created Data Administration of the Projects and Planning Division of the Family History Department should be contacted for approval.

REFERENCES

Genealogical Data Communication (GEDCOM) version dated 9 February 1987

STANDARD

TAG	TAGNAME	DEFINITION
ABY	ABEYANCE	A code associated with the name of an individual to prohibit processing for temple ordinances.
ACTI	ACTION_CODE	Indicates how this data is to affect a file.
ADDI	ADDITIONAL	A relationship or information that exists which gives more detail.
ADDR	ADDRESS	Identifies where a person or business has resided or does or receives mail.
ADMI	ADMINISTRATOR	A person legally vested with the right of administration of an estate.
ADOP	ADOPTION	The act of legally creating a parent-to-child relationship that does not exist by blood.
AENT	A_ENTRIES	Key entry of data by a data entry operator.
AGE	AGE	The age of the individual at the date the document was created or as referenced a document.
ALIA	ALIAS	Alternate name(s) used to identify the same person. Name(s) by which a person is otherwise known.
ALPH	ALPHA_CODE	A control character or set of characters.
ANCE	ANCESTOR	A person from whom other people descended.
AREA	AREA	An LDS ecclesiastical unit presided over by an Area Presidency.
ASSD	ASSIGNED	Responsibility is given to area or department.

TAG	TAGNAME	DEFINITION
ASSI	ASSISTANT	A person who helped or worked with the officiator in performing an ordinance.
BAPL	BAPTISM_LDS	The ordinance of immersion for the remission of sins performed by Latter-day Saint priesthood authority.
BAPM	BAPTISM	The Christian act signifying spiritual rebirth and admitting the recipient to the Christian community through the ritual use of water, often referred to as christening, which is traditionally used.
BATC	BATCH	A group of records processed together.
BENT	B_ENTRIES	The re-keying of data for verification.
BIC	BIC	Born in the Covenant.
BIRT	BIRTH	The event of entering into life.
BLES	BLESSING	The ordnance of bestowing or invoking divine concern, care, intercession, affirmation, guidance, direction, will, etc.
BLSL	BLESSING_LDS	The ordinance of blessing and naming a child.
BRID	BRIDE	Woman to be married.
BUC	BUC	Born under the covenant.
BURI	BURIAL	The event of disposing of the mortal remains of a person who has died.
CANC	CANCELLATION OF_SEALING	A decree by the President of the Church nullifying a sealing ordinance.

TAG	TAGNAME	DEFINITION
CAUS	CAUSE	The reason.
CENS	CENSUS	A periodic count of the population.
CHAN	CHANGES	Corrections and/or modifications to be or that have been made.
CHAR	CHARACTER	This tag is used to identify a change in character set for a transmission.
CHEC	CHECK_SUM	A mathematical procedure used to help verify the accuracy of data.
CHEK	CHECK_BY	The name of the examiner.
CHIL	CHILD	Natural, adopted, or sealed offspring of a father and mother.
CHR	CHRISTENING	The non-LDS ceremony of baptizing and naming.
CITA	CITATION	Reference to the source of the information used for genealogical purposes.
CITY	CITY	An incorporated municipal unit.
CIVI	CIVIL_CONDITION	Marital status of the individual as listed in the census.
CLEA	CLEARED	The status of an individual record which has been approved for proxy temple ordinances
CLRK	CLERK	The person who was the recorder.
CO	COUNTY	A local administrative unit or territorial division in some countries.

TAG	TAGNAME	DEFINITION
CODE	CODE	A symbolic method of representing data or an occurrence.
CODI	CODICIL	An addition, change, or amendment to a will.
COLO	COLOR	A symbol indicating the color of an individual as listed in some US census records.
COMM	COMMENT	Additional information about a specific item or subject.
CONE	CONFIDENTIAL _CODE	A code which indicates the information can be accessed only by authorized individuals. It may refer to a field or an entire record.
CONF	CONFIRMATION	A Christian rite conferring the gift of the Holy Ghost and among protestants full church membership.
CONL	CONFIRMATION_LDS	The ordinance by which a person receives the Gift of the Holy Ghost and becomes a member of the Church.
CONT	CONTINUATION	An indicator that additional information follows.
CORR	CORRECTION	A modification to or update of an existing record and/or field.
COUN	COUNT	Number of items in a batch.
COUP	COUPLE	A husband and his wife.
CREA	CREATION	The process in which data comes into existence in its current environment.

TAG	TAGNAME	DEFINITION
CTRY	COUNTRY	That portion of the locality which identifies the particular country.
DATA	DATA	Stored information.
DATE	DATE	The period in time when an event took place.
DAU	DAUGHTER	A female child described by a relationship to parents.
DEAT	DEATH	The event terminating mortal life.
DESC	DESCENDANT	One who is descended from, such as a child or grandchild.
DEST	DESTINATION	Terminating point of a journey.
DIV	DIVORCE	A civil action dissolving a marriage. Divorce does not terminate a temple sealing.
DUP	DUPLICATE	Recurrence of the same data or information about the sane person.
DWEL	DWELLING	Place of residence.
EMIG	EMIGRATION	The act of leaving a homeland with the intent of locating elsewhere.
END	END	A termination.
ENDL	ENDOWMENT_LDS	One of the essential temple ordinances of the Church required for exaltation.
ENTR	ENTRY	A name or label accompanied by genealogical identifiers that are entered into the computer and given an "entry number."

TAG	TAGNAME	DEFINITION
ENUM	ENUMERATION DISTRICT	A division within an area for census enumeration.
ENUR	ENUMERATOR	Person accumulating the census information.
EOF	END_OF_FILE	An indicator that no more records exist in a file.
EVAL	EVALUATION	An indication that a record has been evaluated.
EVEN	EVENT	The recorded happening used to identify an individual.
EXCO	EXCOMMUNICATION	A recorded event which terminated church membership.
EXEC	EXECUTOR	The person appointed by a testator to execute his will.
EXPL	EXPLANATION	A note giving detail concerning a specific item.
EXTR	EXTRACTOR	The person who a person's name and genealogical information from a record into a standard format for inclusion in computer file.
FAM	FAMILY	A husband and wife and their children, if any.
FAMC	FAMILY_CHILD	A family in which an individual appears as a child.
FAMF	FAMILY_FILE	Records for temple work for which the family will provide proxies.
FAMO	FAMILY_ ORGANIZATION	A group of people who are doing research for a particular family for a particular time period and/or locality.
FAMR	FAMILY_ REPRESENTATIVE	Any individual who is a designated representative for a family.
FAMS	FAMILY_SOURCE	A family in which an individual appears as a spouse.
FATH	FATHER	A male parent.

TAG	TAGNAME	DEFINITION
FEMA	FEMALE	A woman or girl.
FGR	FAMILY_GROUP RECORD	A record that shows a family group consisting of father, mother, and children (if any).
FILE	FILE	A storage place, ordered and arranged for preservation and reference.
FLAG	FLAG	An indicator in a record.
FOLI	FOLIO	An identifying reference.
FONL	FILE_ONLY	A flag indicating that this record contains a Family Group Record form submitted for use in the four-generation program, or donated to the department for use in genealogical research.
FORE	FORENAME	A first name or given name.
FOST	FOSTER	A sealing of a child to foster parents.
FRAM	FRAME	One picture of the series on a roll of microfilm.
GENE	GENEALOGY	The study of ancestors and descendants and their families.
GIVN	GIVEN_NAME	The name or names, excluding the surname, used to identify a person.
GNRL	GENERAL	Reference to a collection of documents in a citation that refer to the entire family listed.
GROO	GROOM	A man who is to be married.